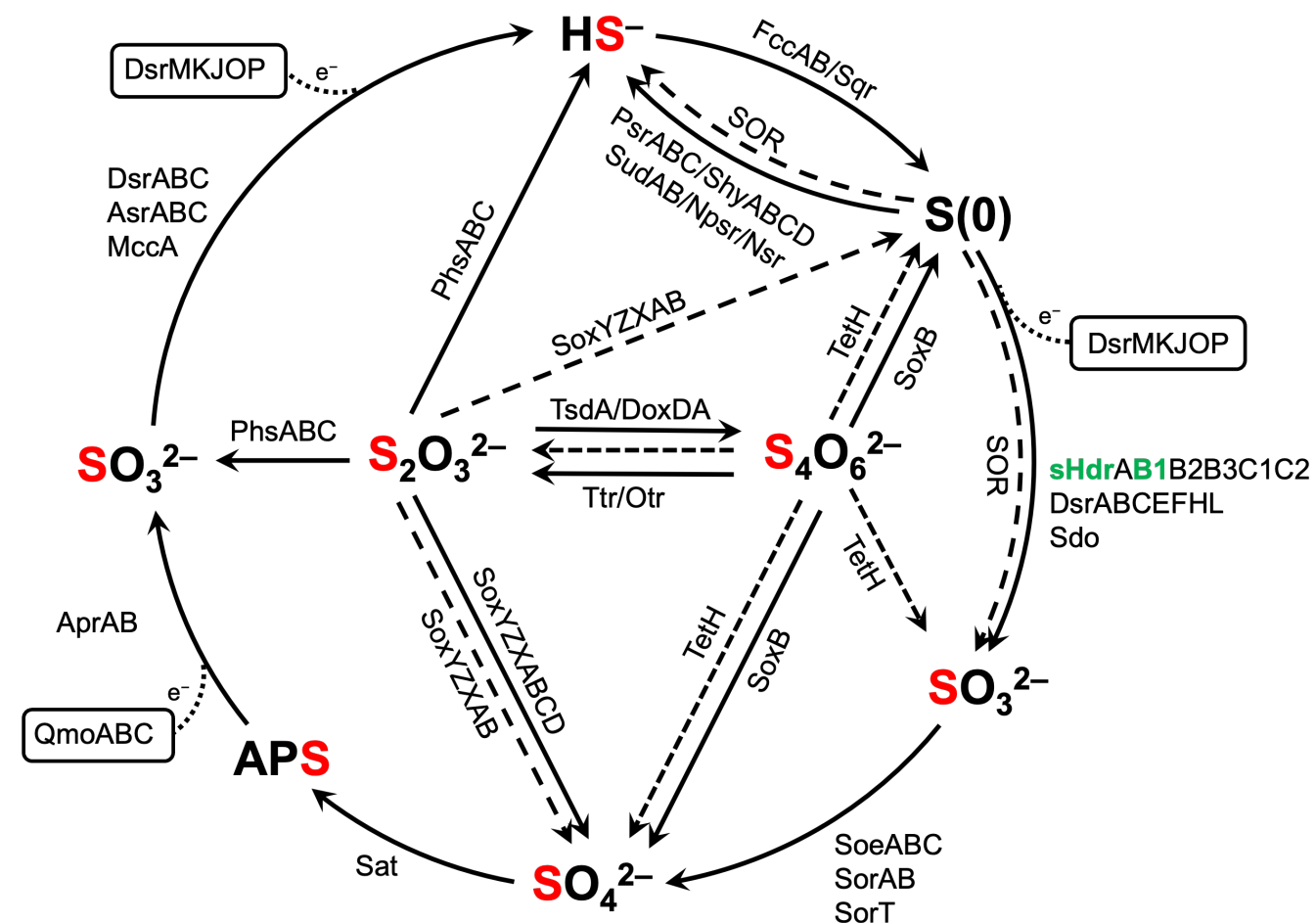


Kumpel et al. (2024)

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Cytoplasm



The diagram illustrates the Sox pathway for sulfur metabolism, showing the conversion of elemental sulfur (S^0) to sulfate (SO_4^{2-}) and the role of various proteins and cofactors.

Top Pathway (Sulfur Oxidation):

- SoxB** (blue oval) catalyzes the oxidation of H_2O to SO_4^{2-} and $2H^+$, releasing $2e^-$.
- SoxYZ** (orange box) is a membrane protein that receives $2e^-$ from **SoxB** and transfers them to **SoxAX** (blue oval), forming a disulfide bond ($S-S$).
- SoxAX** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.
- SoxYZ** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.
- SoxYZ** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.

Bottom Pathway (Sulfur Reduction):

- SoxYZ** (orange box) is reduced by **SoxB** (blue oval) to S^0 , releasing $2e^-$.
- SoxYZ** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.
- SoxYZ** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.
- SoxYZ** is then reduced back to S^0 by **SoxB**, releasing $2e^-$.

Other Pathways:

- TsdA** (yellow box) catalyzes the conversion of $2S_2O_3^{2-}$ to $S_4O_6^{2-}$.
- SorAB** (yellow box) and **SorT** (yellow box) catalyze the conversion of SO_3^{2-} to SO_4^{2-} .
- SoeABC** (green box) and **DoxDA** (green box) are membrane proteins that catalyze the conversion of MQ to MQH_2 .

Legend:

- type I** (green box): **sHdrB1**, **sHdrC1**, **sHdrA**, **sHdrH**
- type II** (blue box): **sHdrB2**, **sHdrC2**, **sHdrA**, **sHdrH**

Diagram Labels:

- Periplasm** (top)
- Membrane** (middle)
- Cytoplasm** (bottom)